

PIZZA

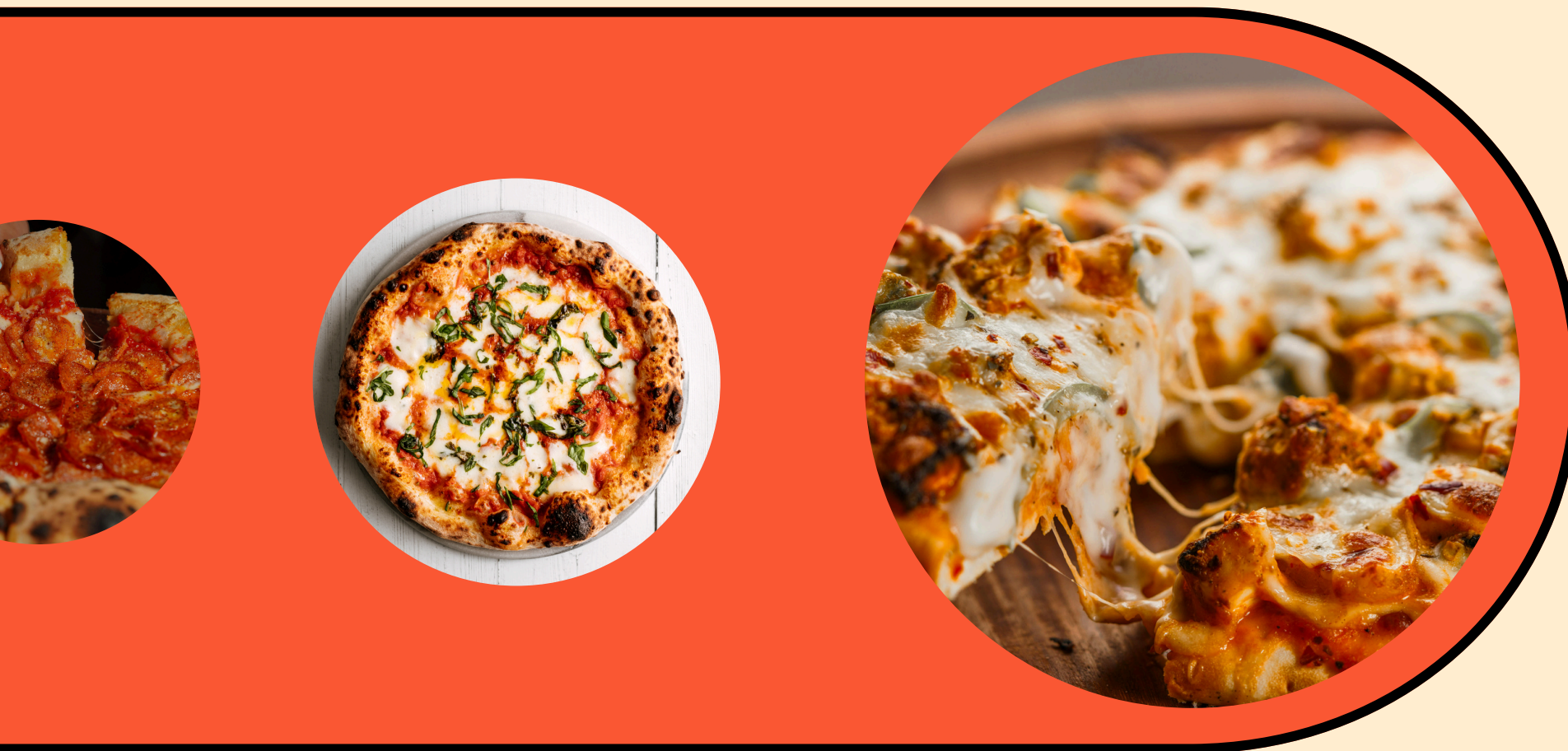
SALES



PIZZA SALES USING MYSQL

This project focuses on analyzing pizza sales data using SQL to extract meaningful business insights. The objective is to transform raw sales data into valuable information that can help improve business decisions, understand customer purchasing behavior, and optimize overall sales performance.

The dataset consists of multiple related tables, including orders, order_details, pizzas, and pizza_types. By applying SQL concepts such as JOINS, GROUP BY, aggregate functions, subqueries, Common Table Expressions (CTEs), and window functions, the project answers important business questions related to sales, revenue, customer demand, and product performance.



OUR QUERIES



Basic:

- Retrieve the total number of orders placed.
- Calculate the total revenue generated from pizza sales.
- Identify the highest-priced pizza.
- Identify the most common pizza size ordered.
- List the top 5 most ordered pizza types along with their quantities.

Intermediate:

- Join the necessary tables to find the total quantity of each pizza ordered.
- Determine the distribution of orders by hour of the day.
- Join relevant tables to find the category-wise distribution of pizzas.
- Group the orders by date and calculate the average number of pizzas ordered per day.
- Determine the top 3 most ordered pizza types based on revenue.

Advanced:

- Calculate the percentage contribution of each pizza type to total revenue.
- Analyze the cumulative revenue generated over time.
- Determine the top 3 most ordered pizza types based on revenue for each pizza category.

RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.



```
SELECT  
    COUNT(order_id) AS total_order  
FROM  
    orders
```

Result Grid	
	total_order
▶	21350



CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

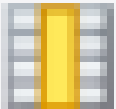
```
SELECT  
    ROUND(SUM(order_details.quantity * pizzas.price),  
          2) AS total_sales  
FROM  
    order_details  
    JOIN  
    pizzas ON pizzas.pizza_id = order_details.pizza_id
```

Result Grid	
	total_sales
▶	817860.05

IDENTIFY THE HIGHEST-PRICED PIZZA.



```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Result Grid   Filter Rows:		
	name	price
▶	The Greek Pizza	35.95

IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.



```
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
    JOIN
        order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC
```

Result Grid			Filter Rows:
	size	order_count	
▶	L	18526	
	M	15385	
	S	14137	
	XL	544	
	XXL	28	

LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.



```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
        order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5
```

Result Grid			Filter Rows:
	name	quantity	
▶	The Classic Deluxe Pizza	2453	
	The Barbecue Chicken Pizza	2432	
	The Hawaiian Pizza	2422	
	The Pepperoni Pizza	2418	
	The Thai Chicken Pizza	2371	

JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA ORDERED.



Result Grid		
	category	quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

```
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC
```

DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.



```
SELECT
    HOUR(order_time) AS hour, COUNT(order_id) AS count
FROM
    orders
GROUP BY HOUR(order_time)
```

Result Grid		
	hour	count
▶	11	1231
	12	2520
	13	2455
	14	1472
	15	1468
	16	1920
	17	2336
	18	2399
	19	2009
	20	1642
	21	1198
	22	663
	23	28

JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.



```
SELECT  
    category, COUNT(name)  
FROM  
    pizza_types  
GROUP BY category
```

	category	COUNT(name)
▶	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9

GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.



```
SELECT
    ROUND(AVG(quantity), 0) AS avg_pizza_order_per_day
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity
```

	avg_pizza_order_per_day
▶	138

DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.



```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

	name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5

CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.



```
SELECT
    pizza_types.category,
    ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
        ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_sales
    FROM
        order_details
        JOIN
            pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100,
        2) AS revenue
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
        order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

	category	revenue
▶	Classic	26.91
	Supreme	25.46
	Chicken	23.96
	Veggie	23.68

ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.



	order_date	cum_revenue
►	2015-01-01	2713.8500000000004
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55
	2015-01-06	14358.5
	2015-01-07	16560.7
	2015-01-08	19399.05
	2015-01-09	21526.4
	2015-01-10	23990.350000000002

```
SELECT order_date,  
       SUM(revenue) OVER (ORDER BY order_date) AS cum_revenue  
FROM (  
  SELECT orders.order_date,  
         SUM(order_details.quantity * pizzas.price) AS revenue  
  FROM order_details  
  JOIN pizzas  
    ON order_details.pizza_id = pizzas.pizza_id  
  JOIN orders  
    ON orders.order_id = order_details.order_id  
  GROUP BY orders.order_date  
) AS sales;
```

DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.



	rn	name	category	revenue
▶	1	The Thai Chicken Pizza	Chicken	43434.25
	2	The Barbecue Chicken Pizza	Chicken	42768
	3	The California Chicken Pizza	Chicken	41409.5
	1	The Classic Deluxe Pizza	Classic	38180.5
	2	The Hawaiian Pizza	Classic	32273.25
	3	The Pepperoni Pizza	Classic	30161.75
	1	The Spicy Italian Pizza	Supreme	34831.25
	2	The Italian Supreme Pizza	Supreme	33476.75
	3	The Sicilian Pizza	Supreme	30940.5
	1	The Four Cheese Pizza	Veggie	32265.70000000065

```
select rn , name , category , revenue from
(select category , name , revenue,
rank() over(partition by category order by revenue desc) as rn
from
(select pizza_types.category , pizza_types.name,
SUM((order_details.quantity )* pizzas.price) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category , pizza_types.name) as a) as b
where rn <=3;
```


PROJECT CONCLUSION

This Pizza Sales Analysis project demonstrates how SQL can be used to transform raw transactional data into meaningful business insights. By applying SQL concepts such as JOINS, aggregate functions, GROUP BY, subqueries, Common Table Expressions (CTEs), and window functions, the project successfully analyzed sales performance, customer ordering patterns, revenue trends, and product popularity.

The analysis identified key metrics such as total revenue, top-selling pizzas, peak ordering hours, category-wise sales distribution, and cumulative revenue growth. These insights can help businesses make informed decisions regarding inventory management, pricing strategies, marketing campaigns, and menu optimization.





THANK YOU